c) The leading strand away from replication fork

d) The lagging strand away from replication fork

28. When two different genes came from same parent they tend to remain together.

    i) What is the name of this phenomenon?

    ii) Draw the cross with suitable example.

    iii) Write the observed phenotypic ratio.

29. What is the difference between missense and nonsense mutation?

30. 

    From the above figure identify the type of mutation and explain it.

31. Write the salient features of Sutton and Boveri concept.

32. Explain the mechanism of crossing over.

33. How is ***Nicotiana*** exhibit self-incompatibility. Explain its mechanism.

34. How sex is determined in monoecious plants. write their genes involved in it.

35. What is gene mapping? Write its uses.

36. Draw the diagram of different types of aneuploidy.

37. Mention the name of man-made cereal. How it is formed?

38. What is DNA repair?

39. What is replication fork?

40. Write about the energetics of DNA replication.

41. What is TATA box?

42. What is alternative splicing?

43. What is coding strand?

44. What are the enzymes involved in DNA replication in eukaryotes?

45. Differentiate coding and non coding strand.

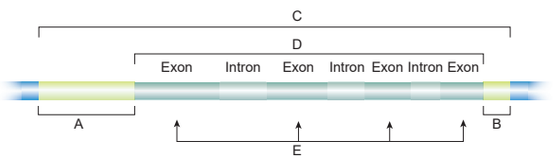
46. What are splicesomes?

47. What is meant by capping and tailing?

48. What is RNA editing?

49. Explain the DNA replication in eukaryotes.

50. With reference to the given diagram correctly match the following pairs.



Column I	Column II
A	Transcribed region
B	Regulation of initiation of transcription
C	Protein-encoding sequence
D	Termination of transcription
E	Gene

51. What attributes make Arabidopsis a suitable model plant for molecular genetic research?

52. Describe the molecular mechanism of RNA modification.

53. Explain ribosomal translocation in protein synthesis.

54. Describe transposons.

55. Describe RNA editing in plants.

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